UNIVERSITY OF WATERLOO

SYDE 556/750 – Simulating Neurobiological Systems

Practice Test 4: Decisions, the LMU, Learning & Representation

Student ID: _____ Time: 30 minutes Total: 20 marks (plus 2 bonus)

- **Closed book.** No notes, textbook, calculator, phone, or other devices.
- The marks for each question are shown in brackets on the right.
- Write your answers in the space provided. Show your work — partial credit is given for correct reasoning.
- Unless stated otherwise, every synapse is the first-order low-pass $h(t) = \frac{1}{\tau} e^{-t/\tau}$, and a recurrent population with transforms A', B' on synapse τ implements $\dot{x} = Ax + Bu$ with $A' = \tau A + I$, $B' = \tau B$.
- PES uses Nengo's sign convention: error = actual − target.

Question 1. **[4 marks]**

A 1-D ensemble with radius 1 has a recurrent connection computing the identity with `transform=1` on a $\tau = 0.2$ s synapse. A constant input u enters through `transform=1` on a $\tau_{\text{in}} = 20$ ms synapse. A second, 1-D ensemble receives the decoded function choice$(x) = 1$ if $x > 0.9$, -1 if $x < -0.9$, and 0 otherwise.

(a) Give A and B for the system the first ensemble implements. For $u = 0.1$ switched on at $t = 0$, find when x first reaches 0.9, first ignoring the input-synapse mismatch, and then including the offset $(1 - \tau_{\text{in}}/\tau)\, u$ that it adds.

(b) In the box, sketch the accumulator and the decision output for $u = 0.1$ and for $u = -0.05$ over $0 \le t \le 4$ s. Label the times at which each decision is made. Give two ways in which the decoded decision output differs from the requested choice(x).

Question 2. **[4 marks]**

A Legendre Memory Unit with $q = 3$ and window $\theta = 0.25$ s is built in one 3-D ensemble, with $\tau = 0.05$ s on both the recurrent and the input connection. Its matrices are

$$\theta A = \begin{pmatrix} -1 & -1 & -1 \\ 3 & -3 & -3 \\ -5 & 5 & -5 \end{pmatrix}, \qquad \theta B = \begin{pmatrix} 1 \\ -3 \\ 5 \end{pmatrix}.$$

The window is read out as $u(t-\theta') \approx \sum_{i=0}^{2} x_i(t)\, \tilde{P}_i(\theta'/\theta)$, with $\tilde{P}_0(r) = 1$, $\tilde{P}_1(r) = 2r - 1$, $\tilde{P}_2(r) = 6r^2 - 6r + 1$.

(a) Give the recurrent transform A' and the input transform B', with their shapes.

(b) At some instant the state is $x = (0.5,\ 0.3,\ -0.2)$. Write the stored window as a polynomial in $r = \theta'/\theta$. Give the current input, the input θ ago, and the average input over the window. In the box, sketch $u(t-\theta')$ against θ' for $0 \le \theta' \le 0.25$ s.

Question 3. **[4 marks]**

(a) A 2-D post population learns with PES, $\Delta w_{ij} = -\eta\,\alpha_i \langle e_i, \varepsilon \rangle a_j$, with $\eta = 10^{-4}$. A pre neuron fires at $a_j = 50$ Hz and the current error is $\varepsilon = (0.1,\ -0.2)$. Post neuron 1 has $e_1 = (0.6,\ 0.8)$ and $\alpha_1 = 2$; post neuron 2 has $e_2 = (0.8,\ -0.6)$ and $\alpha_2 = 2$. Find Δw_{1j} and Δw_{2j}, and explain in one sentence why their signs move the decoded output toward the target.

(b) A PES connection ($\eta = 10^{-4}$, initialized to output 0) learns $f(x) = x$ from a band-limited white-noise input with RMS 0.5. At $t = 10$ s the error population is switched to use the target $f(x) = -x$. In the box, sketch the RMSE between the output and the current target over $0 \le t \le 20$ s. Give its approximate value just after $t = 0$ and just after $t = 10$ s, with reasons.

Question 4. **[4 marks]**

Three neurons with rates $a_1 = 1 + x$, $a_2 = 1 - x$ and $a_3 = 1$ are sampled at $x = -1, 0, 1$. Let $\mathbf{A}$ be the 3×3 matrix with one row per sample and one column per neuron.

(a) Compute $\gamma = \mathbf{A}^\top \mathbf{A}$. Show that $(1, 1, 1)$, $(1, -1, 0)$ and $(1, 1, -2)$ are eigenvectors, and give the singular values. Find $\chi = \mathbf{AU}$ and describe each column as a function of x.

(b) What does the zero singular value mean for this population? Write every decoder $\mathbf{D}$ that decodes $f(x) = x$ exactly, and say which one the SVD pseudo-inverse returns. With activity noise $\sigma^2 = 4$, by what factor is the decoded $\hat{x}$ shrunk?

Question 5. **[4 marks]**

A pendulum obeys $\ddot{\theta} = \tau_{\text{applied}} - mgL\cos\theta$ with $mgL = 9.8$ (and $mL^2 = 1$). The angle is measured from the horizontal. A PD controller applies $u_{\text{PD}} = 16(\theta^* - \theta) - 8\dot{\theta}$.

(a) With PD control only, write the condition for the steady angle. Estimate it for $\theta^* = 0$ using $\cos\theta \approx 1$, and say whether the exact value is larger or smaller in magnitude. What is the steady angle for $\theta^* = \pi/2$, and why?

(b) An adaptive ensemble representing $(\theta, \dot{\theta})$ adds its decoded output u_{ad} to the torque, through a PES connection initialized to 0. What signal (with sign) drives its learning rule, and what function does it learn? Give its value at $\theta = 0$ and at $\theta = \pi/2$. In the box, sketch $\theta(t)$ with and without adaptation while the target alternates between 0 and $\pi/2$ every 5 s, starting at 0, over 40 s.

Bonus Question 1. **[2 bonus marks]**

A synapse follows STDP, $\Delta w = \sum_{\text{pairs}} W(t_{\text{post}} - t_{\text{pre}})$, with $W(\Delta t) = \alpha^{+} e^{-\Delta t/10\,\text{ms}}$ for $\Delta t > 0$ and $W(\Delta t) = -\alpha^{-} e^{\Delta t/10\,\text{ms}}$ for $\Delta t \le 0$. The pre neuron spikes at 0 and 20 ms; the post neuron spikes at 5 and 15 ms. Find the total Δw over all four pairs when $\alpha^{+} = \alpha^{-}$, and when $\alpha^{+} = 1$, $\alpha^{-} = 0.5$. Why does plain rate-based Hebbian learning, $\Delta w = \eta a_{\text{pre}} a_{\text{post}}$, not distinguish these two spike orders?